

Terna Engineering College

Computer Engineering Department

Class: TE

Sem.: VI

Course: System Security Lab

PART A

(PART A : TO BE REFFERED BY STUDENTS)

Experiment No.9

A.1 Aim: Simulate DOS attack using Hping, hping3 and Wireshark

A.2 Prerequisite:

1. Basic Knowledge of DOS attacks,

A.3 Outcome:

After successful completion of this experiment students will be able to

To be able to use open source technologies and explore email security and explore various attacks.

A.4 Theory:

Denial-of-service (DoS) attack is an attempt to make a machine or network resource unavailable to its intended users, such as to temporarily or indefinitely interrupt or suspend services. A distributed denial-of-service (DDoS) is where the attack source is more than one, often thousands of, unique IP addresses. It is analogous to a group of people crowding the entry door or gate to a shop or business, and not letting legitimate parties enter into the shop or business, disrupting normal operations.

A DoS attack tries to make a web resource unavailable to its users by flooding the target URL with more requests than the server can handle. That means that during the attack period, regular trafficon the website will be either slowed down or completelyinterrupted.

A Distributed Denial of Service (DDoS) attack is a DoS attack that comes from more than one source at the same time. A DDoS attack is typically generated using thousands (potentially hundreds of thousands) of unsuspecting zombie machines. The machines used in such attacks are collectively known as “botnets” and will have previously been infected with malicious software, so they can be remotely controlled by the attacker. According to research, tens of millions of computers are likely to be infected with botnet programs worldwide.

Cybercriminals use DoS attacks to extort money from companies that rely on their websites being accessible. But there have also been examples of legitimate businesses having paid underground elements of the Internet to help them cripple rival websites. In addition, cybercriminals combine DoS attacks and phishing to target online bank customers. They use a DoS attack to take down the bank's website and then send out phishing e-mails to direct customers to a fake emergency site instead.

Installation Steps:

1. Install Hping3 and Wireshark
2. Flood the victim with TCP/ICMP/UDP packet using Hping3 (-- flood option)
3. Observe the DoS attack and DDoS attack using Wireshark

PART B

(PART B : TO BE COMPLETED BY STUDENTS)

(Students must submit the soft copy as per following segments within two hours of the practical. The soft copy must be uploaded on the Blackboard or emailed to the concerned lab in charge faculties at the end of the practical in case the there is no Black board access available)

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Class : TE-C	Batch : C3
Date of Experiment:	Date of Submission: 28-3-25
Grade :	

B.1 Output

(add snapshot of output)

```
Starting Nping 0.7.95 ( https://nmap.org/nping ) at 2025-03-28 17:14 IST
SENT (0.0046s) TCP 127.0.0.1:60449 > 127.0.0.1:80 5 ttl=64 id=35008 iplen=40 seq=794605948 win=1480
RCVD (0.0046s) TCP 127.0.0.1:80 > 127.0.0.1:60449 RA ttl=64 id=11592 iplen=40 seq=0 win=0
SENT (0.0146s) TCP 127.0.0.1:60449 > 127.0.0.1:80 5 ttl=64 id=35008 iplen=40 seq=794605948 win=1480
RCVD (0.0146s) TCP 127.0.0.1:80 > 127.0.0.1:60449 RA ttl=64 id=16198 iplen=40 seq=0 win=0
SENT (0.0246s) TCP 127.0.0.1:60449 > 127.0.0.1:80 5 ttl=64 id=35008 iplen=40 seq=794605948 win=1480
RCVD (0.0247s) TCP 127.0.0.1:80 > 127.0.0.1:60449 RA ttl=64 id=41687 iplen=40 seq=0 win=0
SENT (0.0347s) TCP 127.0.0.1:60449 > 127.0.0.1:80 5 ttl=64 id=35008 iplen=40 seq=794605948 win=1480
RCVD (0.0348s) TCP 127.0.0.1:80 > 127.0.0.1:60449 RA ttl=64 id=50406 iplen=40 seq=0 win=0
SENT (0.0448s) TCP 127.0.0.1:60449 > 127.0.0.1:80 5 ttl=64 id=35008 iplen=40 seq=794605948 win=1480
RCVD (0.0449s) TCP 127.0.0.1:80 > 127.0.0.1:60449 RA ttl=64 id=53418 iplen=40 seq=0 win=0

Max rtt: 0.077ms | Min rtt: 0.037ms | Avg rtt: 0.059ms
Raw packets sent: 5 (200%) | Rcvd: 5 (200%) | Lost: 0 (0.00%)
Nping done: 1 IP address pinged in 0.04 seconds

+ -- sudo nping --tcp -p 80 --rate 100 127.0.0.1

Starting Nping 0.7.95 ( https://nmap.org/nping ) at 2025-03-28 17:15 IST
SENT (0.0062s) TCP 127.0.0.1:53845 > 127.0.0.1:80 5 ttl=64 id=41812 iplen=40 seq=1686822046 win=1480
RCVD (0.0063s) TCP 127.0.0.1:80 > 127.0.0.1:53845 RA ttl=64 id=41558 iplen=40 seq=0 win=0
SENT (0.0163s) TCP 127.0.0.1:53845 > 127.0.0.1:80 5 ttl=64 id=41812 iplen=40 seq=1686822046 win=1480
RCVD (0.0164s) TCP 127.0.0.1:80 > 127.0.0.1:53845 RA ttl=64 id=19830 iplen=40 seq=0 win=0
SENT (0.0263s) TCP 127.0.0.1:53845 > 127.0.0.1:80 5 ttl=64 id=41812 iplen=40 seq=1686822046 win=1480
RCVD (0.0264s) TCP 127.0.0.1:80 > 127.0.0.1:53845 RA ttl=64 id=22178 iplen=40 seq=0 win=0
SENT (0.0364s) TCP 127.0.0.1:53845 > 127.0.0.1:80 5 ttl=64 id=41812 iplen=40 seq=1686822046 win=1480
RCVD (0.0365s) TCP 127.0.0.1:80 > 127.0.0.1:53845 RA ttl=64 id=16120 iplen=40 seq=0 win=0
SENT (0.0466s) TCP 127.0.0.1:53845 > 127.0.0.1:80 5 ttl=64 id=41812 iplen=40 seq=1686822046 win=1480
RCVD (0.0467s) TCP 127.0.0.1:80 > 127.0.0.1:53845 RA ttl=64 id=13085 iplen=40 seq=0 win=0

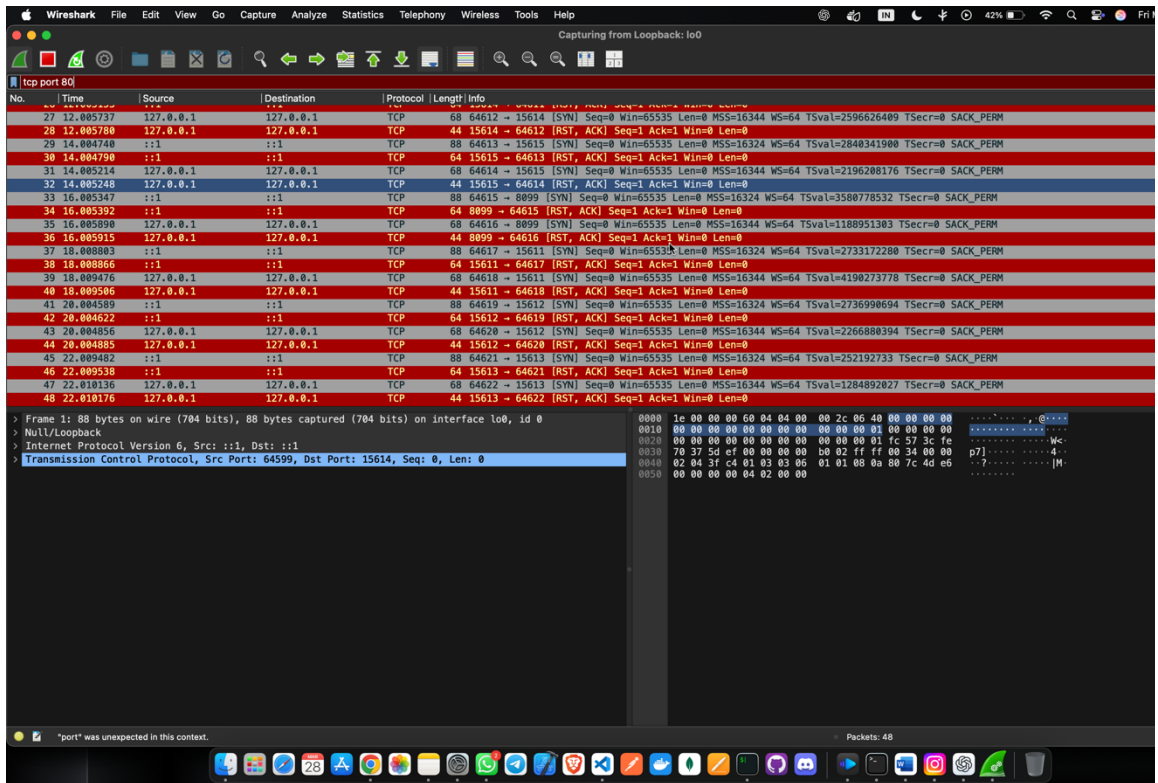
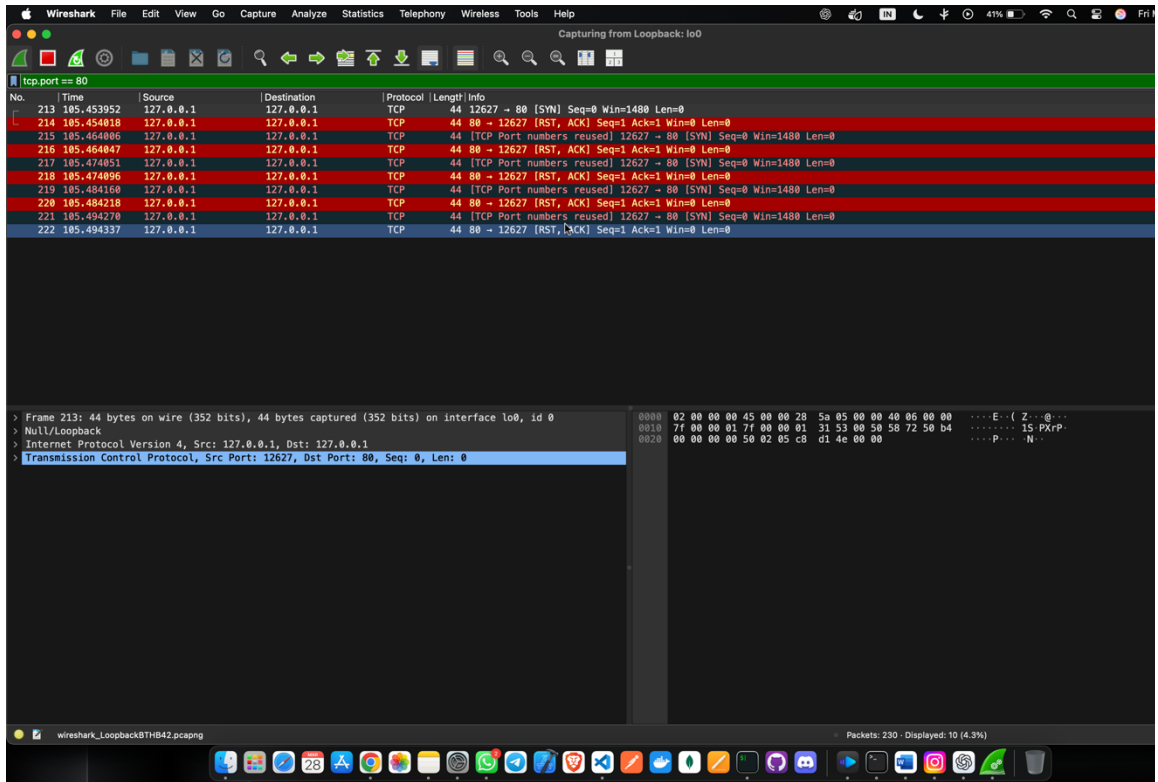
Max rtt: 0.080ms | Min rtt: 0.064ms | Avg rtt: 0.070ms
Raw packets sent: 5 (200%) | Rcvd: 5 (200%) | Lost: 0 (0.00%)
Nping done: 1 IP address pinged in 0.05 seconds

+ -- sudo nping --tcp -p 80 --rate 100 127.0.0.1

Starting Nping 0.7.95 ( https://nmap.org/nping ) at 2025-03-28 17:18 IST
SENT (0.0066s) TCP 127.0.0.1:12627 > 127.0.0.1:80 5 ttl=64 id=23045 iplen=40 seq=1483886772 win=1480
RCVD (0.0067s) TCP 127.0.0.1:80 > 127.0.0.1:12627 RA ttl=64 id=38323 iplen=40 seq=0 win=0
SENT (0.0167s) TCP 127.0.0.1:12627 > 127.0.0.1:80 5 ttl=64 id=23045 iplen=40 seq=1483886772 win=1480
RCVD (0.0167s) TCP 127.0.0.1:80 > 127.0.0.1:12627 RA ttl=64 id=25266 iplen=40 seq=0 win=0
SENT (0.0267s) TCP 127.0.0.1:12627 > 127.0.0.1:80 5 ttl=64 id=23045 iplen=40 seq=1483886772 win=1480
RCVD (0.0268s) TCP 127.0.0.1:80 > 127.0.0.1:12627 RA ttl=64 id=58056 iplen=40 seq=0 win=0
SENT (0.0368s) TCP 127.0.0.1:12627 > 127.0.0.1:80 5 ttl=64 id=23045 iplen=40 seq=1483886772 win=1480
RCVD (0.0369s) TCP 127.0.0.1:80 > 127.0.0.1:12627 RA ttl=64 id=45604 iplen=40 seq=0 win=0
SENT (0.0469s) TCP 127.0.0.1:12627 > 127.0.0.1:80 5 ttl=64 id=23045 iplen=40 seq=1483886772 win=1480
RCVD (0.0470s) TCP 127.0.0.1:80 > 127.0.0.1:12627 RA ttl=64 id=23646 iplen=40 seq=0 win=0

Max rtt: 0.080ms | Min rtt: 0.053ms | Avg rtt: 0.065ms
Raw packets sent: 5 (200%) | Rcvd: 5 (200%) | Lost: 0 (0.00%)
Nping done: 1 IP address pinged in 0.05 seconds

+ --
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B.2 Commands / tools used with syntax:

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sudo nping --tcp -p 80 --rate 100 127.0.0.1
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B.3 Question of Curiosity: *(Attempt at least 3 questions handwritten)*

Q1] What is the difference between DoS & DDoS?

DoS	DDoS
i) Attack on single system	Attack on Multiple system
ii) Limited impact	Long-scale impact
iii) Easier to detect & block	Harder to build due to multiple sources
iv) eg. SYN Flood, Ping of Death	eg. Botnet Band flooding, Mirai Botnet attack

Q2] What is ping of death attack?

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- i) The Ping of death is a type of denial of service attack
 - ii) It involves sending malformed or oversized ICMP packets to a target system
 - iii) The target system may crash, freeze or rebooted & to buffer overflow.

Q3] What is land Attack?

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- i) A land Attack is a type of DoS attack that exploits vulnerabilities in how a system handles network packets
 - ii) The attacker sends a spoofed SYN packet where both source & destination IP addresses are the same
 - iii) The victim system becomes confused as it keeps replying to itself, causing CPU exhaustion & system failure

B.4 Conclusion:

(Write appropriate conclusion.)

In this experiment, we demonstrated how to simulate a Denial-of-Service (DoS) attack using hping3 while analyzing the network traffic using Wireshark. By crafting and sending a large number of TCP, UDP packets to a target system, we observed how excessive traffic can overload network resources, potentially making a system unresponsive.